



International Currency Technologies Inc.

ICT106 General Protocol For RS232

Communication Specification

Change History

[illegible]

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1. General Information

This specification describes the operation of ICT106 Interface Protocol.
It is regarding to the data for the interface between Controller and ICT product.

2. Communication Format

2.1 Transmission Specification

Transmission Method	Full Duplex Transmission	
Transmission Speed	9600 Baud Rate	
Synchronizing Method	Asynchronous Mode	
Connection Control Method	Request Method	
Data Format	Start bit	1
	Data bit	8
	Parity bit	Even
	Stop bit	1
	X parameter	None

2.2 Timing Specification

Response Time (maximum)	50 ms
Polling Time Interval	100 ~ 200 ms

Definition:

Response Time	The maximum time from send out a valid message to receive a response of this message.
Polling Time Interval	The time interval between two message polls.

3. Commands

3.1 Power Up

Bill Acceptor			Controller	
Category	Command	Direction	Command	Description
Power Supply ON	80H (128) 8FH (143)	→		
1. Received 02H in 2 seconds		←	02H (02)	Within 2 seconds

* Bill Acceptor sends “Power Supply ON” Command(80H 8FH) every 2 seconds until receive 0x02 from Controller.

* Bill Acceptor will be Enable status until receive 0x02 from Controller.

3.2 Escrow

Bill Acceptor			Controller	
Category	Command	Direction	Command	Description
Send bill validated	81H (129) 8FH (129)	→		Bill validated
Send bill value	40H ~ 44H	→		40H : First Bill Type 41H : Second Bill Type 42H : Third Bill Type 43H : Fourth Bill Type 44H : Fifth Bill Type
	Wait for Request (5 Seconds)	←	02H (02) Or 0FH (15) Or 18H(24) Or	a. Request to Accept Or b. Request to Reject Or c. *Request to Hold in Escrow d. *No response (5 seconds)
Stacking (for a.) Or Reject (for b)	10H (16) 11H (17)	→		Controller do not need to respond

*Note 1: 18H will hold the bill in Escrow until receive Accept (0x02h) or Reject (0x0fh).

*Note 2: If no any response, after 5 seconds Escrow Timeout, Bill Acceptor will Reject Bill automatically. If it continue 3 Times Escrow Timeout, Bill Acceptor Response Communication Failure.

3.3 Controller Polling Bill Acceptor Status

Bill Acceptor			Controller	
Category	Command	Direction	Command	Description
		←	0CH (12)	Request Bill Acceptor Status
	20H (32) 21H (33) 22H (34) 23H (35) 24H (36) 25H (37) 27H (39) 28H (40) 29H (41) 2AH (42) • • 2EH(46) 2FH(47) 3EH (62) 5EH (94) 71H (113) A1H (161)			Motor Failure Checksum Error Bill Jam Bill Remove Stacker Open Sensor Problem Bill Fish Stacker Problem Bill Reject Invalid Command • • Reserved Response when Error Status is Exclusion Bill Acceptor Enable Status Bill Acceptor Inhibit Status Bill Acceptor busy Power Supply on Communication

* Controller need not to respond

3.4 Enable/Disable

Bill Acceptor			Controller	
Category	Command	Direction	Command	Description
		←	3EH (62)	Controller Enable Bill Acceptor
Response to Enable Bill Acceptor	3EH (62)	→		
		←	5EH (94)	Controller Disable Bill Acceptor
Response to Disable Bill Acceptor	5EH (94)	→		

3.5 Reset

Bill Acceptor			Controller	
Category	Command	Direction	Command	Description
Command Bill Acceptor Reset		←	30H (48)	
Power Supply ON	80H (128) 8FH (143)	→		
1. Received 02H in 2 seconds		←	02H (02)	Within 2 seconds

* Bill Acceptor sends “Power Supply ON” Command(80H 8FH) every 2 seconds until receive 02H from Controller.

* Bill Acceptor will be Enable status until receive 0x02 from Controller.

3.6 Stacking action caused by bill jam.

Bill Acceptor			Controller	
Category	Command	Direction	Command	Description
Stacking Finish cased by bill jam	83H (131)	➔		
Denomination been stacked	3FH ~ 4FH	➔		40H : 1st Bill Type 41H : 2nd Bill Type 42H : 3rd Bill Type 43H : 4th Bill Type 44H : 5th Bill Type . . 4FH : 16th Bill Type 3FH: Unknown Bill Type

* Controller need not to respond

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